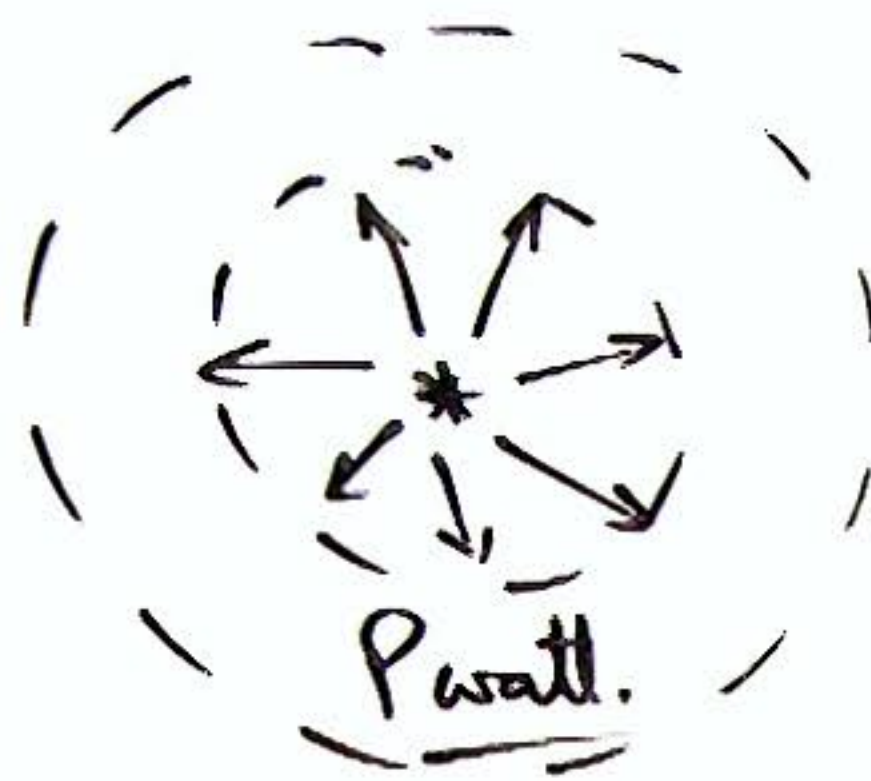


① No. of photons emitted
by a light source :



$$W/L = \lambda$$

$$\text{Energy of one photon} = \frac{hc}{\lambda}$$

$$\text{No of ph/sec } N = \frac{P}{hc/\lambda} = \boxed{\frac{P\lambda}{hc}}$$

② Photon flux in a light beam.



$$\text{photon flux } \phi_p = \frac{N}{S} = \frac{P\lambda}{hcS} = \boxed{\frac{P\lambda}{hc}}$$

③ Photon Density in a Light Beam:

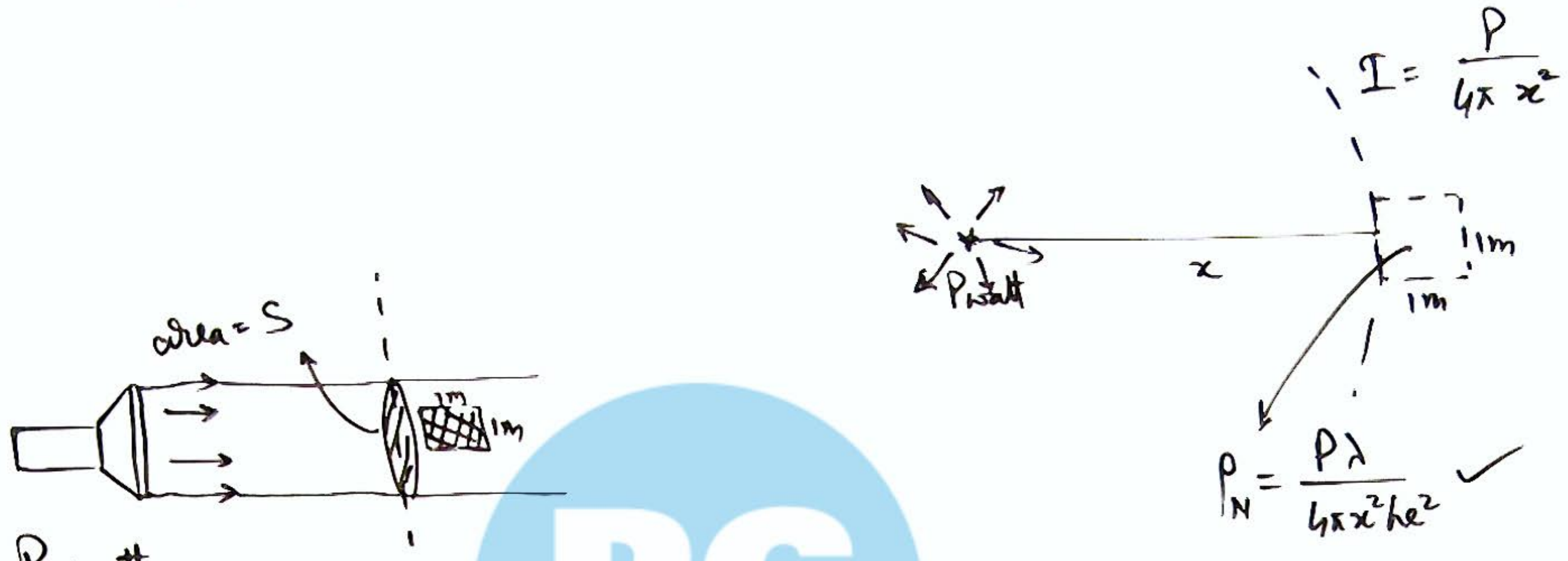


Diagram illustrating photon density in a light beam. A light beam with power P watt passes through an area S . A small area of $1\text{m} \times 1\text{m}$ is shown. The photon flux ϕ_P is given by:

$$\phi_P = \frac{I\lambda}{hc} = \frac{P\lambda}{hcS}$$

Photon density in a light beam is given by:

$$\rho_N = \frac{\phi_P}{c} = \frac{I\lambda}{hc^2}$$

For a point source emitting power P watt at a distance x , the intensity I is:

$$I = \frac{P}{4\pi x^2}$$

The photon density ρ_N at distance x is:

$$\rho_N = \frac{P\lambda}{4\pi x^2 hc^2} \checkmark$$

④ Wave Particle Duality :

Nature of light

- Particle Nature $\rightarrow$ photon (it has momentum) (p)
- Wave Nature $\rightarrow$ wave length of EM wave (λ)

By Relativity theory, energy & momentum of 'light photons' are related as

$$E^2 = p^2 c^2 + m_0^2 c^4 \quad \text{for any particle}$$

for photons

rest mass = 0 $\Rightarrow$ for photons $E^2 = p^2 c^2$

$$\Rightarrow E = \rho_c = \frac{h c}{\lambda} \Rightarrow$$

$$\rho = \frac{h}{\lambda}$$

⑤ De-Broglie's Hypothesis:

for light $\lambda = \frac{h}{p_{\text{photons}}}$

Assumption for physical
particle's wavelength as

$$\lambda = \frac{h}{p_{\text{particle}}}$$

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⑥ Radiation Pressure:

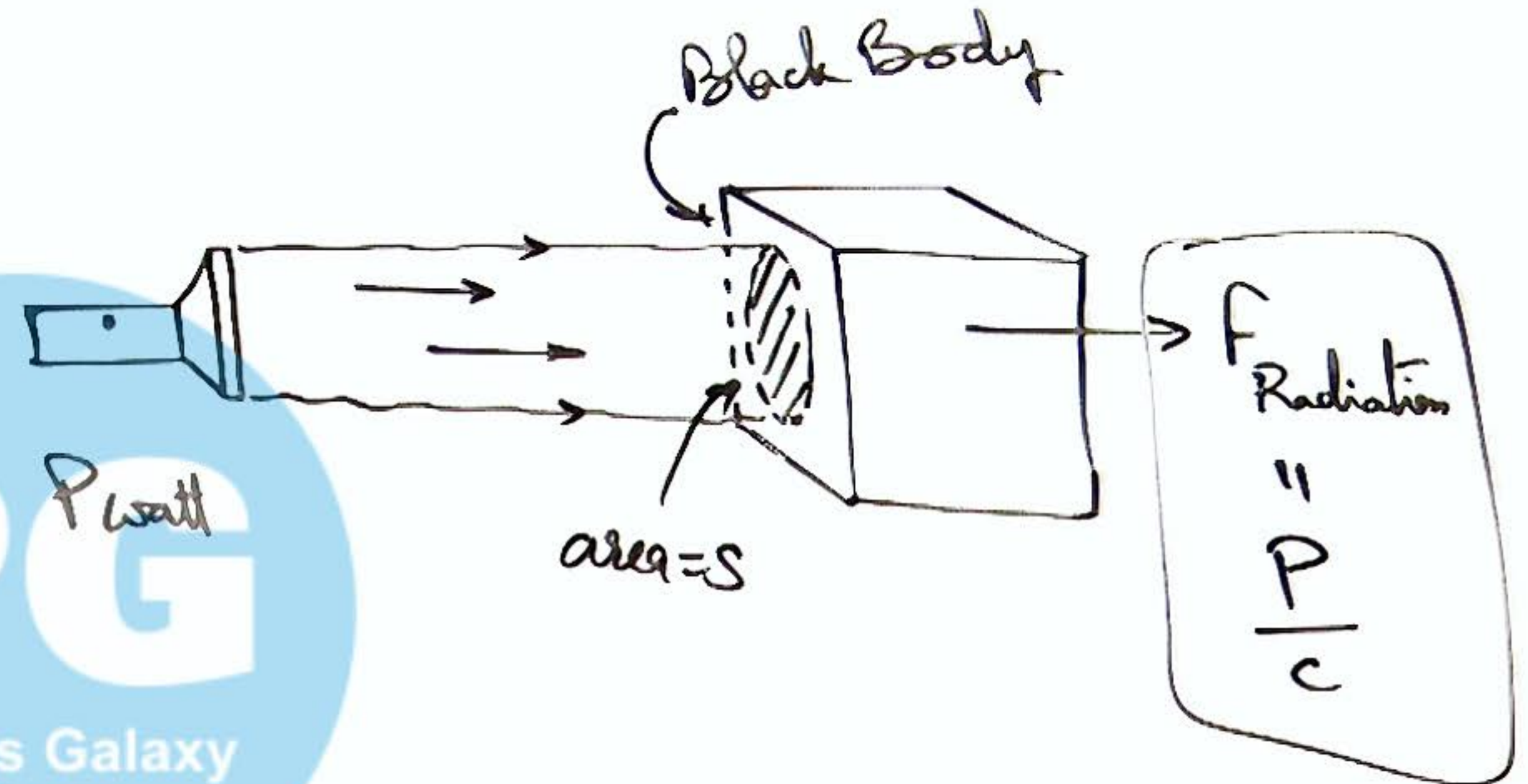
$\lambda/L = \lambda$

 photon
 $p = \frac{h}{\lambda}$

Rad. force on body

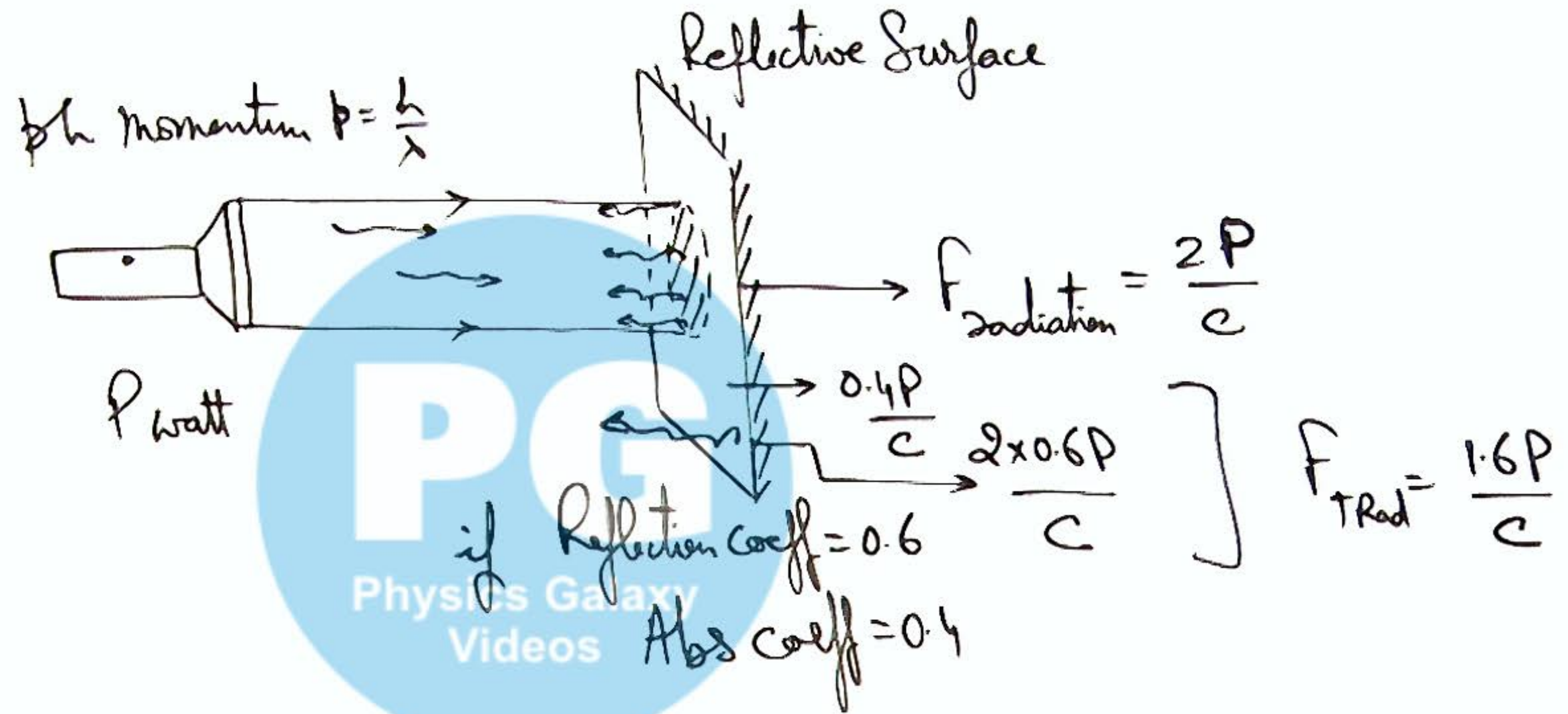
$F = \text{no of ph/sec} \times \text{momentum of one ph.}$

$F = \frac{P \lambda}{hc} \times \frac{h}{\lambda} = \frac{P}{c}$

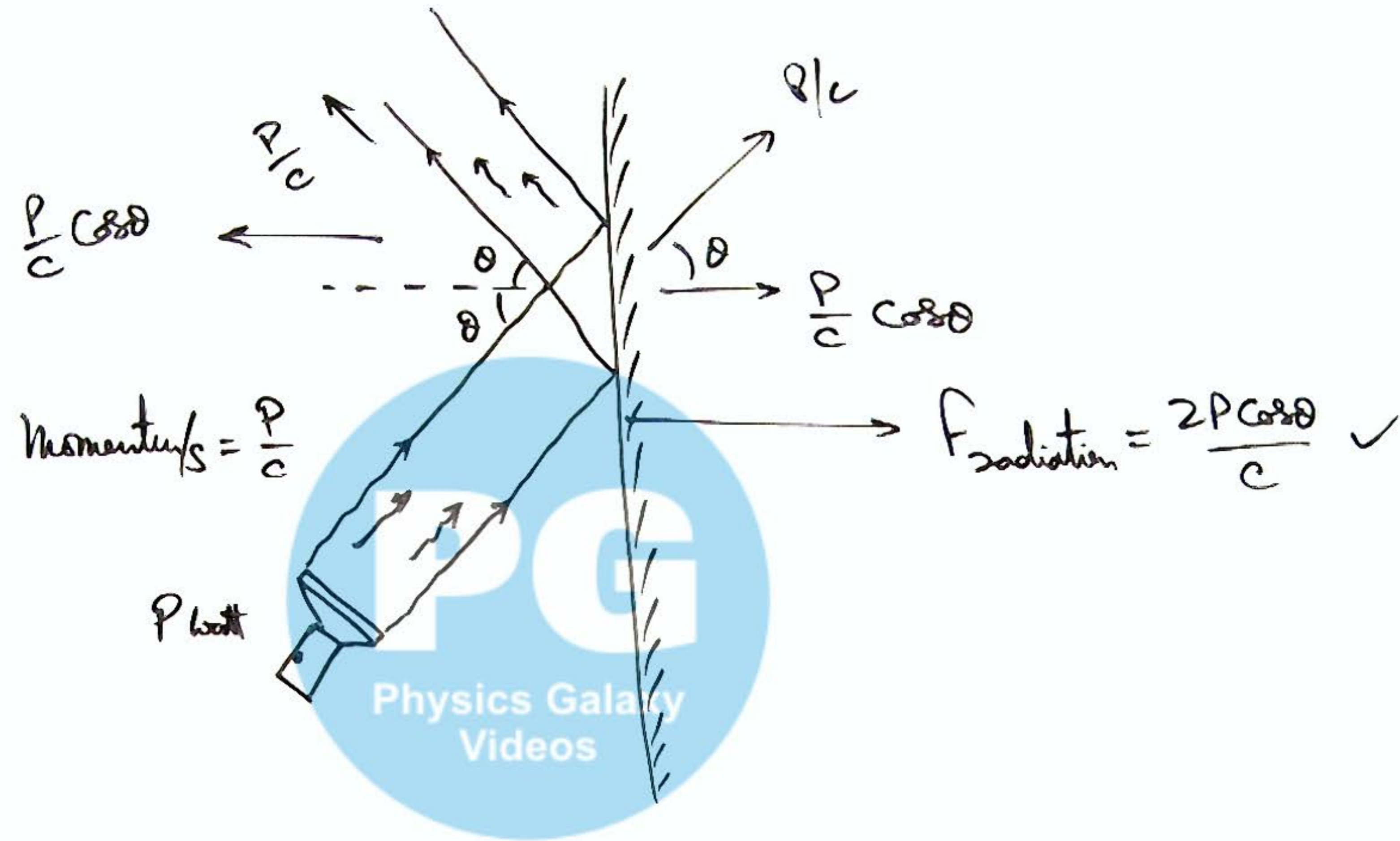


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⑦ Radiation force on a Reflective Surface:

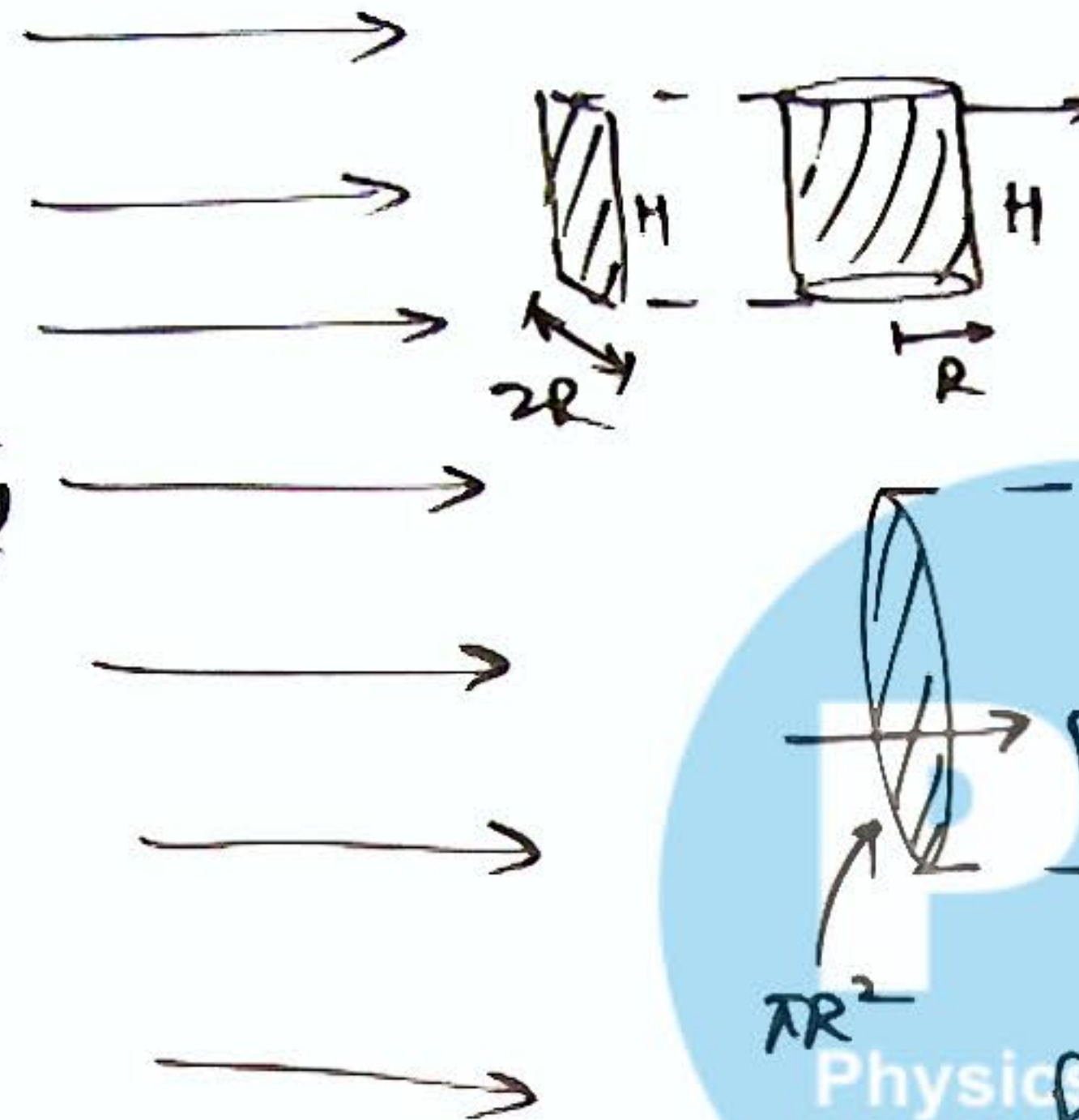


⑧ Radiation Force due to
Oblique Incidence :



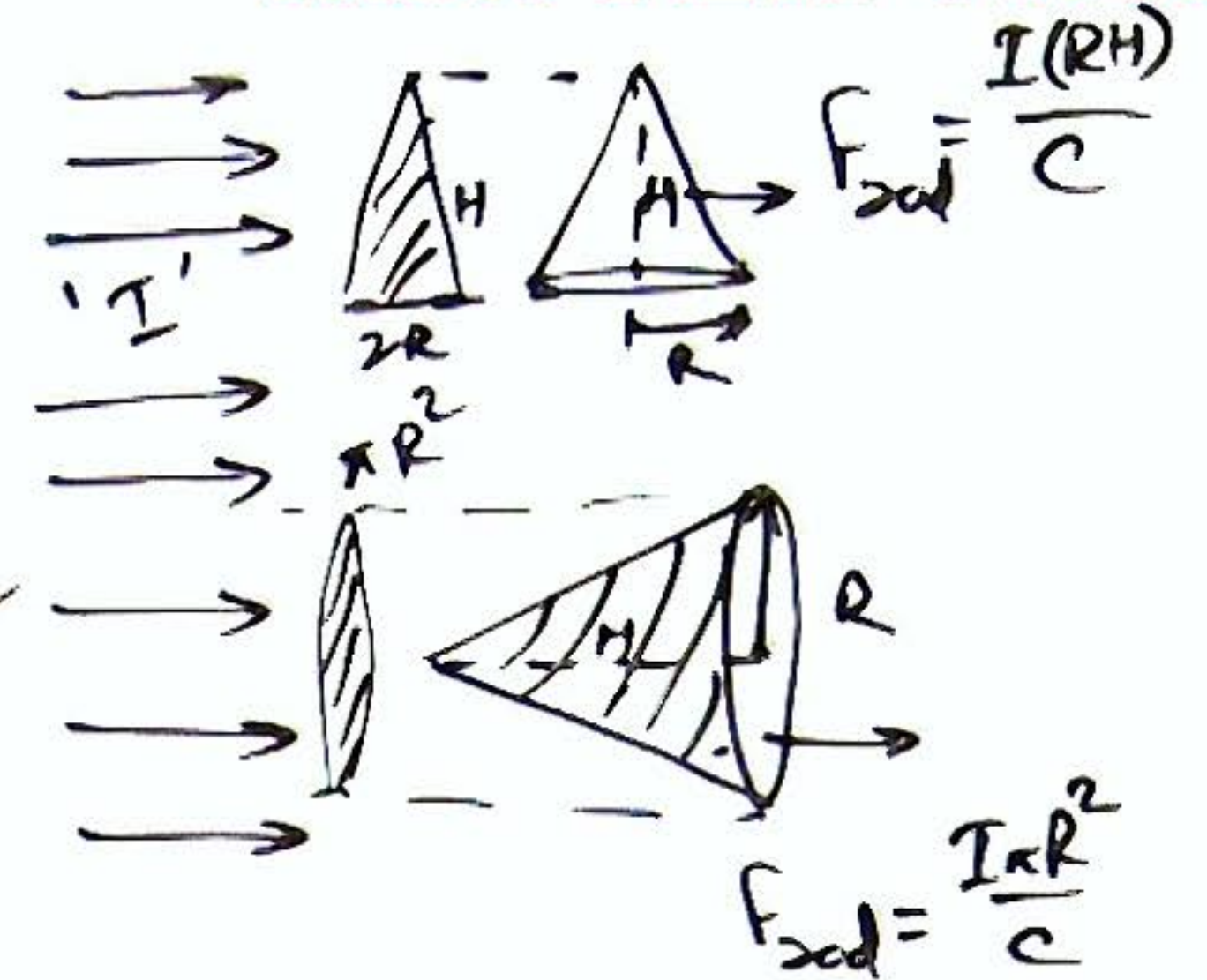
⑨ Radiation Force on objects in path of a light beam:

Intensity
'I'



$$F_{\text{rad}} = \frac{I(2RH)}{c}$$

BB Sphere



Power incident on Sphere $P = I \cdot \pi R^2$

$$F_{\text{rad}} = \frac{P}{c} = \frac{I \pi R^2}{c}$$

imp NOTE: The force on "Sphere" is same for all types of surfaces

⑩ Recoiling of an Atom
during de-excitation :

Initially at rest

Recoil vel.
 V_R

$p_R = mV_R$

By cons of linear momentum

✓ $mV_R = \frac{h}{\lambda}$

$\Rightarrow V_R = \frac{h}{m\lambda}$

$n=4$
 $\downarrow e^-$
 $n=2$

photon

$\Delta E = 2.55 \text{ eV}$

$\lambda = \frac{12431}{2.55} \text{ \AA}$

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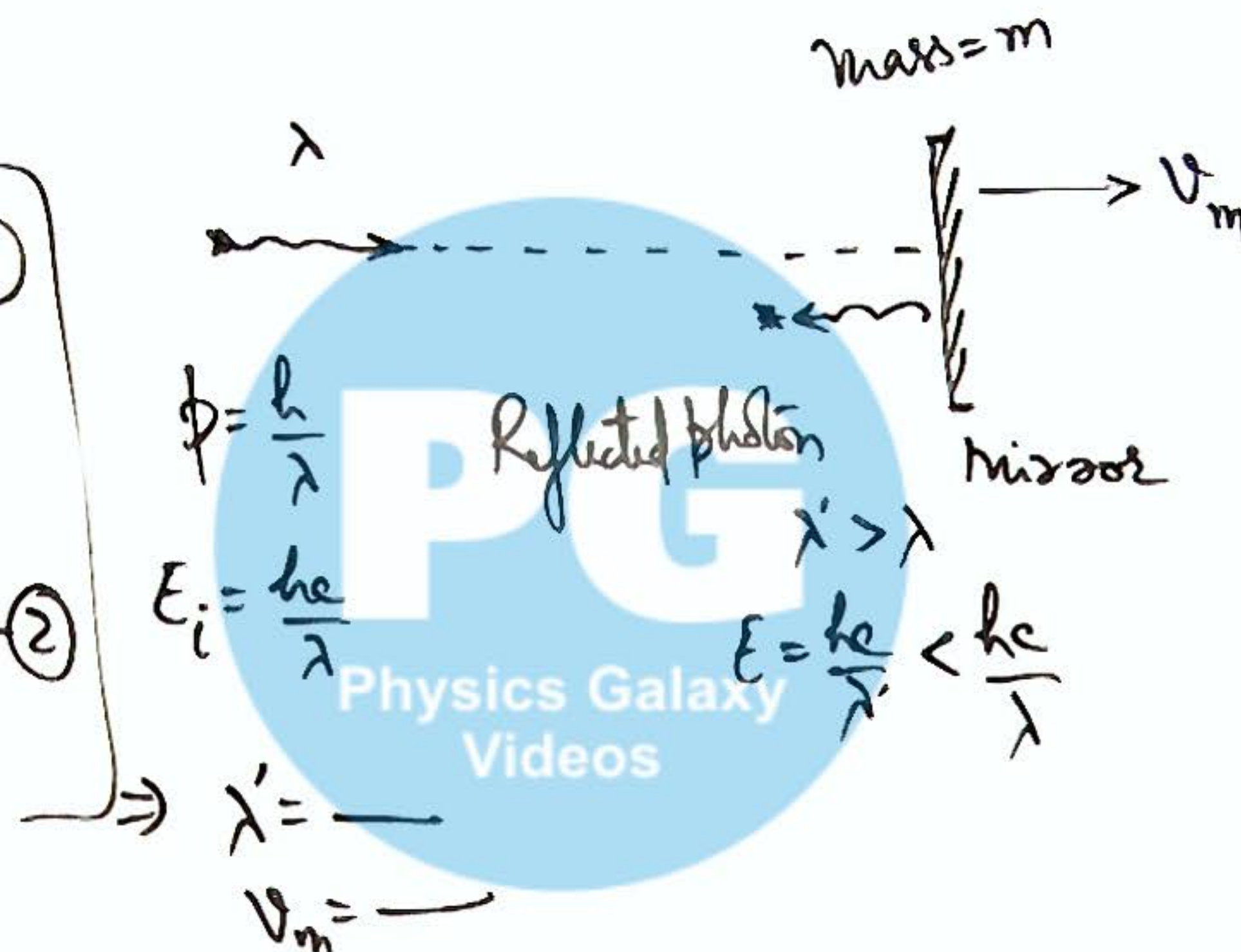
⑪ Change in Wavelength due to Reflection of light :

by momentum Cons

$$\frac{h}{\lambda} = mv_m - \frac{h}{\lambda'} \quad \text{--- (1)}$$

by Energy Conservation

$$\frac{hc}{\lambda} = \frac{hc}{\lambda'} + \frac{1}{2}mv_m^2 \quad \text{--- (2)}$$



$$p_m = mv_m$$

$$K_m = \frac{1}{2}mv_m^2$$